

AAYUSH NITIN MAHESHWARKAR

Senior Software Engineer · Java · Spring Boot · Microservices · AWS · Distributed Systems

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PROFESSIONAL SUMMARY

Senior Software Engineer with 5+ years architecting Java/Spring Boot microservices that handle 2.5M+ peak daily requests and contribute to \$600M+ annualized revenue. Hands-on with distributed systems, caching, OAuth/JWT security, resilience patterns, and event-driven architectures. Production experience with LLM instruction engineering and agentic AI pipelines using LangChain and Claude. AWS Solutions Architect (in progress).

EXPERIENCE

Senior Software Engineer · Federal Express Corporation

July 2021 – Present

Orlando, FL · Promotion: Full Stack Developer II → III (Feb 2023)

- **\$600M+ revenue impact:** Architected enterprise Java/Spring Boot rating microservices serving 2.5M+ peak daily rate requests across the global rating platform.
- **Quote Platform — \$450M billed / \$1.8B quoted across 69 markets:** Led modernization of backend rating logic in Java and Spring Boot to support spot pricing outside contractual rates; FY23 contribution drove FedEx Hall of Fame recognition (FY24).
- **Surcharge calculation engine — \$150M+ annualized revenue:** Built core Java/Spring Boot engine exposed via REST APIs for high-throughput surcharge assessment; integrated into the 2.5M req/day rating pipeline.
- **Shipping eligibility microservices — \$41M direct + \$1.2B linked revenue:** Owned end-to-end design and delivery of dangerous-goods routing using Java/Spring Boot, Feign clients, and a configurable rules engine; enabled migration of 34,000+ enterprise customers across regulated, country-specific shipping rules.
- **~600ms p99 latency reduction on customer-facing rate-quote API:** Designed proactive Okta JWT pre-caching using Caffeine refreshAfterWrite with async refresh, eliminating token-fetch overhead from the critical rating path under 2.5M+ daily requests; pattern adopted by adjacent teams.
- **Production AI governance — Microsoft Copilot:** Owned quality standards and final-approval authority for LLM instruction sets across four enterprise platforms; defined prompt evaluation criteria for clarity, completeness, security, and accuracy, gating git commits on benchmark adherence.
- **Agentic AI pipeline — LangChain + Claude:** Contributed to a LangChain and Claude-powered agentic pipeline built to handle a projected 10x surge in error-correction volume from 2,000 to 20,000 daily shipments, eliminating a 5-day manual remediation cycle. Implemented Resilience4j retry/backoff and authored Python automation for bulk data rectification.
- **Cache Optimization:** Implemented Caffeine Loading Cache across REBS services by identifying ideal cache candidates to maintain pre-loaded cache state, reducing downstream service calls and improving rating pipeline performance.

Teaching Assistant · Rensselaer Polytechnic Institute

Aug 2019 – Dec 2020

Computer Organization · Intro to Formal Logic and AI · Troy, NY

- Mentored 300+ students across Computer Organization (C, MIPS, x86 assembly) and Intro to Formal Logic and AI; conducted code reviews and supported proof construction in HyperSlate.

TECHNICAL SKILLS

Languages: Java (17/21), Python, SQL, Shell, C

Backend & Microservices: Spring Boot, Spring Cloud, Spring Security, Hibernate/JPA, REST, Feign, Resilience4j, JMS, Kafka (familiar); Service Discovery (Eureka), Circuit Breakers, API Gateway, Saga, CQRS, Idempotency, Event-driven architecture

Cloud / DevOps: AWS (EC2, S3, Lambda, API Gateway, RDS, SQS, SNS, CloudWatch, IAM), Docker, Kubernetes (basic), Jenkins CI/CD, Gradle, Linux, Git

Security & Data: OAuth 2.0, SSO, Okta, JWT, Spring Security; PostgreSQL, Oracle, MySQL, MongoDB, Redis, Caffeine

Observability: Splunk, Dynatrace, Grafana, AppDynamics

AI / LLMs: Agentic AI System Design, LangChain, LLM Instruction Engineering, Prompt Evaluation, Microsoft Copilot, Claude, RAG fundamentals

ACHIEVEMENTS

- **FedEx Hall of Fame Award (FY24):** Quote Platform modernization — \$450M billed / \$1.8B quoted across 69 global markets.
- **AAAI 2020 Fall Symposium on AI for Social Good (Co-Author):** “Socioeconomic and Geographic Variations Impacting the Spread of Malaria.”

EDUCATION

M.S. Information Technology

Aug 2019 – Dec 2020

Rensselaer Polytechnic Institute, Troy, NY | CGPA: 3.8/4.0

B.Tech Electronics and Communication Engineering (Minor: Computer Science)

Aug 2015 – Jul 2019

PES University, Bengaluru, India | CGPA: 7.63/10 | Minor CGPA: 8.95/10